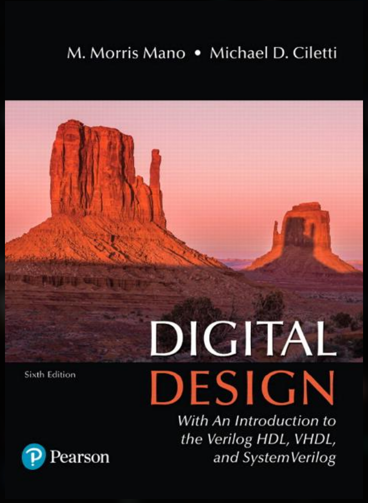
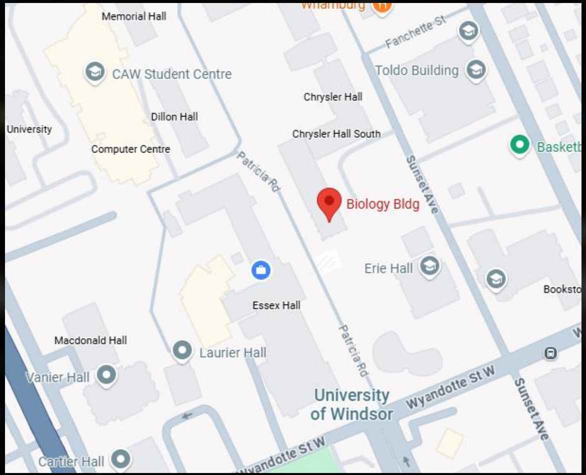


Chapter 1
Digital Systems and Binary Numbers


Office Hour
1:30PM-2:30PM
Lambton Tower 5111

Lab03/Lec03 Extended for Another Week



Final Exam (Tentative)
12-Aug, 8:30 AM
Biology Building (BB) 113

SIGNED MAGNITUDE

SIGNED COMPLEMENT

r^{n-1}	r^{n-2}	r^{n-3}	…	r^2	r^1	r^0
0	Positive Numbers					
Nonzero	Negative Numbers					

Signed

Magnitude

SIGNED MAGNITUDE ARITHMETIC

0	X	+	0	Y	=	0	X+Y
0	X	+	1	Y	=	$X<Y$ (if borrow)	X–Y (if borrow, apply it)
1	X	+	1	Y	=	1	–(X+Y) = X+Y

0	X	–	0	Y	=	$X<Y$ (if borrow)	X–Y (if borrow, apply it)
0	X	–	1	Y	=	0	X+Y
1	X	–	1	Y	=	$X>Y$ (if borrow)	–X+Y=Y–X (if borrow, apply it)

SIGNED MAGNITUDE

OVERFLOW

Overflow!

0	X	+	0	Y	=	$0 \rightarrow 1$	X+Y
0	X	+	1	Y	=	X<Y	X−Y
1	X	+	1	Y	=	$1 \rightarrow 1$	X+Y

0	X	−	0	Y	=	X<Y	X−Y
0	X	−	1	Y	=	$0 \rightarrow 1$	X+Y
1	X	−	1	Y	=	X>Y	−X+Y=Y−X

WHY NOT
SIGNED MAGNITUDE

Give up left most position for sign! What are the wastes?

r^{n-1}	r^{n-2}	r^{n-3}	…	r^2	r^1	r^0
0	Positive Numbers					
Nonzero	Negative Numbers					

$$+0 \rightarrow \text{Max} = r^{n-1} - 1 = \cancel{r^n - 1}$$

$$\text{Min} = -(r^{n-1} - 1) \leftarrow -0$$

SIGNED COMPLEMENT

SIGNED COMPLEMENT

DIMINISHED RADIX COMPLEMENT

Given $(N)_r$ with n digits, the (r – 1)'s complement of N, i.e., its *diminished radix complement*, is defined as $(r^n - 1) - N$.

Distance to the Largest Number (Largest Digits in Each Position)

1's COMP. BASE-2

Base-2		2^4	2^3	2^2	2^1	2^0
$2^5 =$	1	0	0	0	0	0

Base-2		2^4	2^3	2^2	2^1	2^0
$2^5=$	1	0	0	0	0	0
$1=$		0	0	0	0	1

Base-2		2^4	2^3	2^2	2^1	2^0
		+2	+2	+2	+2	
	–1	–1	–1	–1	–1	+2
$2^5=$	1	0	0	0	0	0
$1=$		0	0	0	0	1
$2^5–1=$		1	1	1	1	1
	5 digits of 1					

Base-2	2^5	2^4	2^3	2^2	2^1	2^0
$2^5–1=$		1	1	1	1	1
N=		1	0	1	0	1
$(2^5–1)–N=$		0	1	0	1	0
1's complement of $(10101)_2=(01010)_2=$ NOT on each digit						

3's COMP. BASE-4

Base-4		4^4	4^3	4^2	4^1	4^0
$4^5 =$	1	0	0	0	0	0

Base-4		4^4	4^3	4^2	4^1	4^0
$4^5=$	1	0	0	0	0	0
$1=$	0	0	0	0	0	1

Base-4		4^4	4^3	4^2	4^1	4^0
		+4	+4	+4	+4	
	–1	–1	–1	–1	–1	+4
$4^5=$	1	0	0	0	0	0
$1=$		0	0	0	0	1
$4^5–1=$		3	3	3	3	3
	5 digits of 3					

Base-4		4^4	4^3	4^2	4^1	4^0
$4^5 - 1 =$		3	3	3	3	3
N=		1	2	1	3	0
$(4^5 - 1) - N =$		2	1	2	0	3

3's complement of $(12130)_4 = (21203)_4 =$ 3 – Each digit

9's COMP. BASE-10

Base-10		10^4	10^3	10^2	10^1	10^0
		$+10$	$+10$	$+10$	$+10$	
	-1	-1	-1	-1	-1	$+10$
$10^5 =$	1	0	0	0	0	0
$1 =$		0	0	0	0	1
$10^5 - 1 =$		9	9	9	9	9
	5 digits of 9					

Base-10		10^4	10^3	10^2	10^1	10^0
$10^5-1=$		9	9	9	9	9
$N=$		1	2	1	3	0
$(10^5-1)-N=$		8	7	8	6	9
9's complement of $(12130)_{10}=(87869)_{10}=$ 9 – Each digit						

(r–1)’s COMP. BASE-r

Base-r		r^{n-1}	…	r^2	r^1	r^0
$r^n-1=$		$r-1$	…	$r-1$	$r-1$	$r-1$
$N=$		d_{n-1}	…	d_2	d_1	d_0
$(r^n-1)-N=$		$r-1-d_{n-1}$	…	$r-1-d_{n-1}$	$r-1-d_{n-1}$	$r-1-d_{n-1}$
$(r-1)$'s complement of $(N)_r=(r-1)$ – Each digit						

RADIX COMPLEMENT

Given $(N)_r$ with n digits, the r's complement of N, i.e., its *radix complement*, is defined as $r^n - N$.

RADIX COMPLEMENT

Given $(N)_r$ with n digits, the r's complement of N, i.e., its *radix complement*, is defined as $r^n - N$.

Diminished Complement + 1

$(r-1)$'s Complement + 1 = $[(r^n - 1) - N] + 1 = r^n - N$

2’s COMP. BASE-2

Base-2		2^4	2^3	2^2	2^1	2^0
$2^5-1=$		1	1	1	1	1
$N=$		1	0	1	0	1
$(2^5-1)-N=$		0	1	0	1	0
NOT on each digit						
$+1=$		0	0	0	0	1
$2^5-N=$		0	1	0	1	1
2's complement of $(10101)_2=(01010+1)_2=(01011)_2$						

4's COMP. BASE-4

Base-4		4^4	4^3	4^2	4^1	4^0
$4^5 – 1=$		3	3	3	3	3
N=		1	2	1	3	0
$(4^5 – 1) – N=$		2	1	2	0	3
3 – Each digit						
+1=		0	0	0	0	1
$4^5 – N=$		2	1	2	1	0

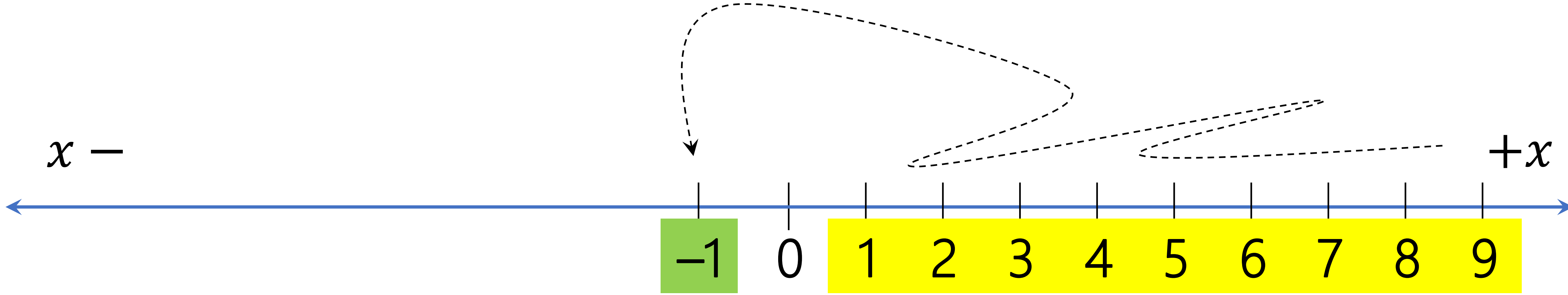
4's complement of $(12130)_4 = (21203+1)_4 = (21210)_4$

10's COMP. BASE-10

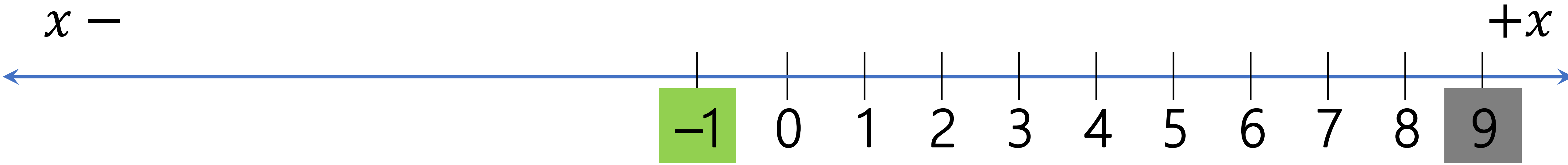
Base-10		10^4	10^3	10^2	10^1	10^0
$10^5-1=$		9	9	9	9	9
$N=$		1	2	1	3	0
$(10^5-1)-N=$		8	7	8	6	9
9 – Each digit						
$+1=$		0	0	0	0	1
$10^5-N=$		8	7	8	7	0
10's complement of $(12130)_{10}=(87869+1)_{10}=(87870)_{10}$						

r’s COMP. BASE-r

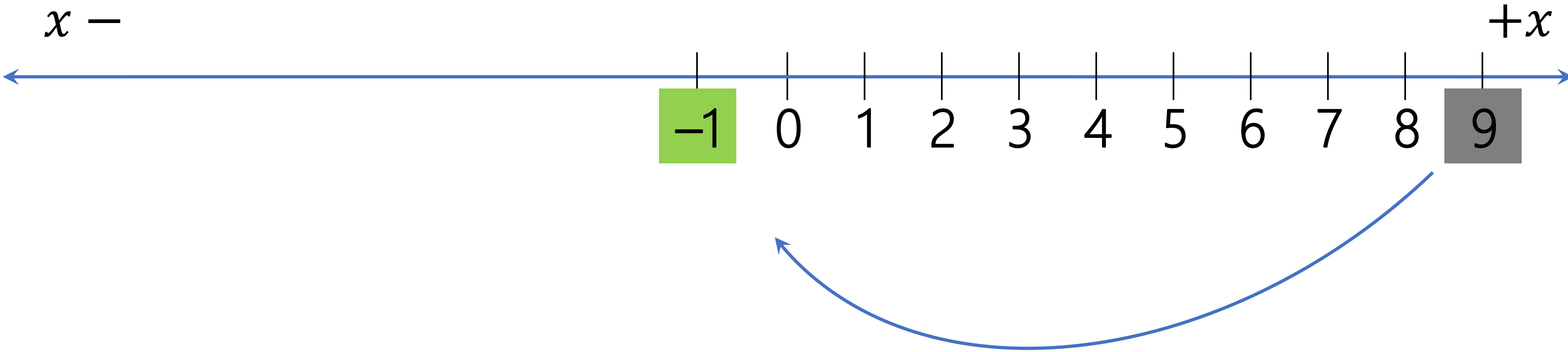
SIGNED COMPLEMENT



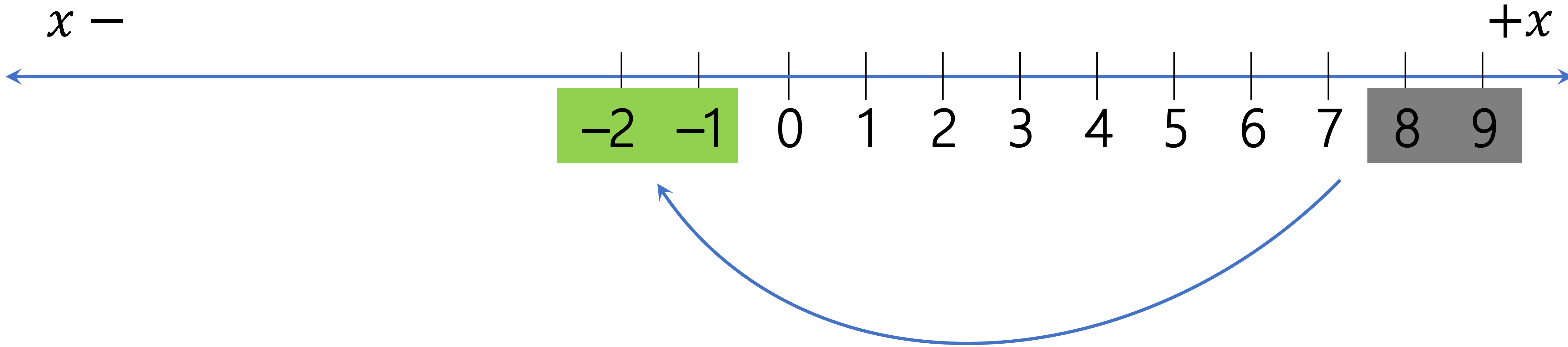
Which one do you scarify?



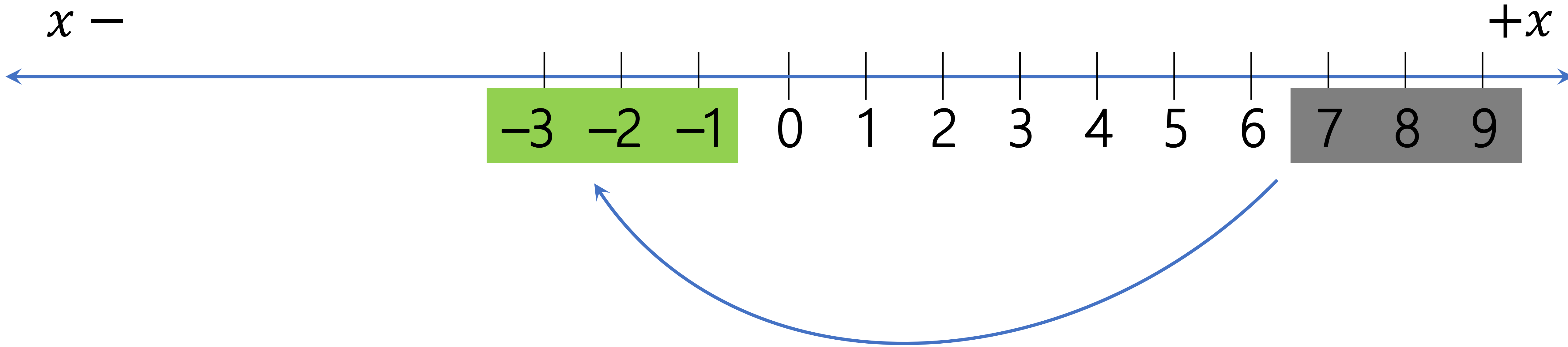
x −
+x
−1
0
1
2
3
4
5
6
7
8
9



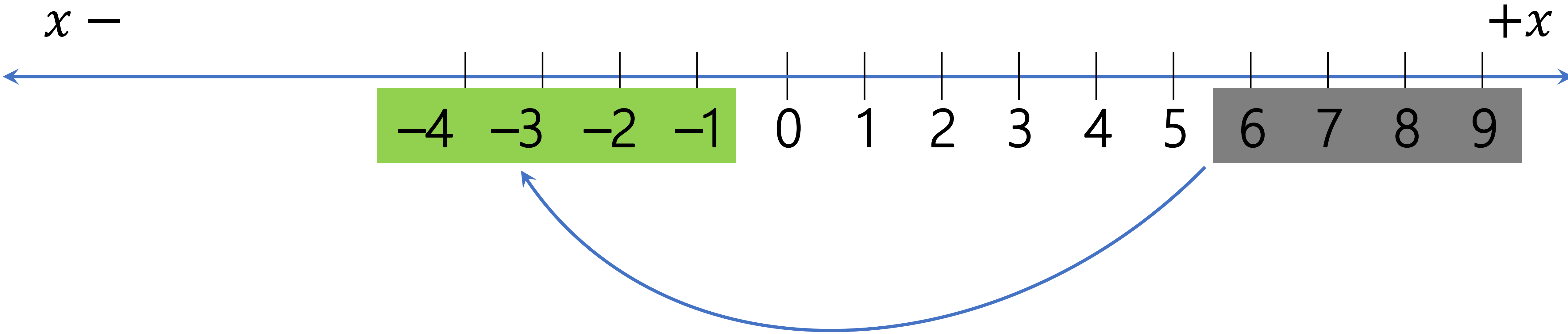
$$-1 = -(10^1 - 9) = -(\text{10's comp. } 9)$$



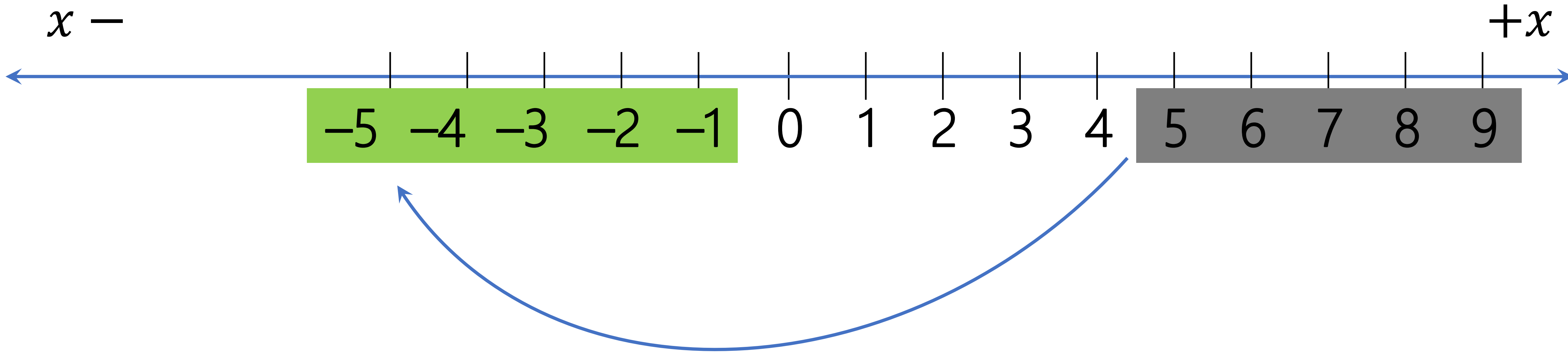
$$-2 = -(10^1 - 8) = -(10\text{'s comp. } 8)$$



$$-3 = -(10^1 - 7) = -(10\text{'s comp. } 7)$$

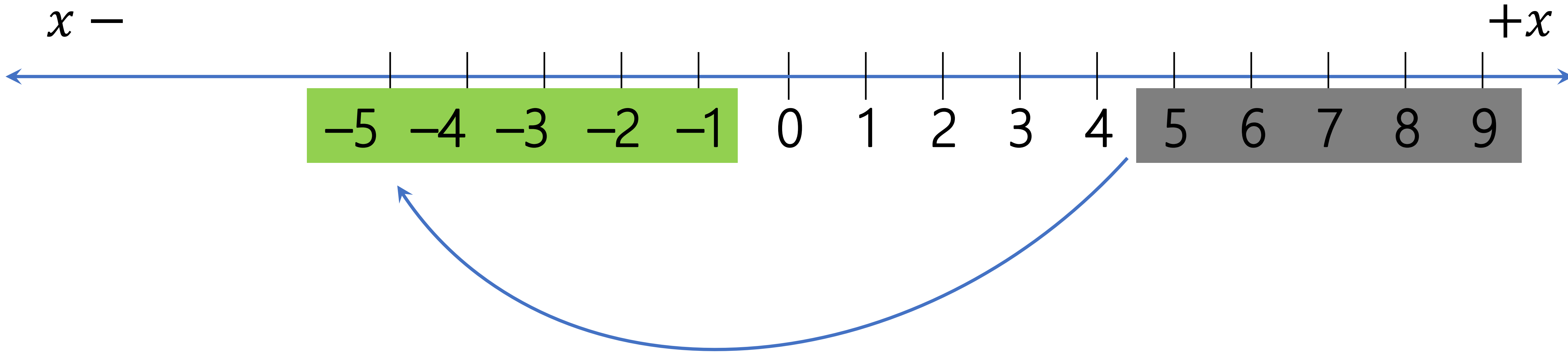


$$-4 = -(10^{1} - 6) = -(10\text{'s comp. } 6)$$



$$-5 = -(10^1 - 5) = -(10\text{'s comp. } 5)$$

Which one you keep/drop, 5 or -5?



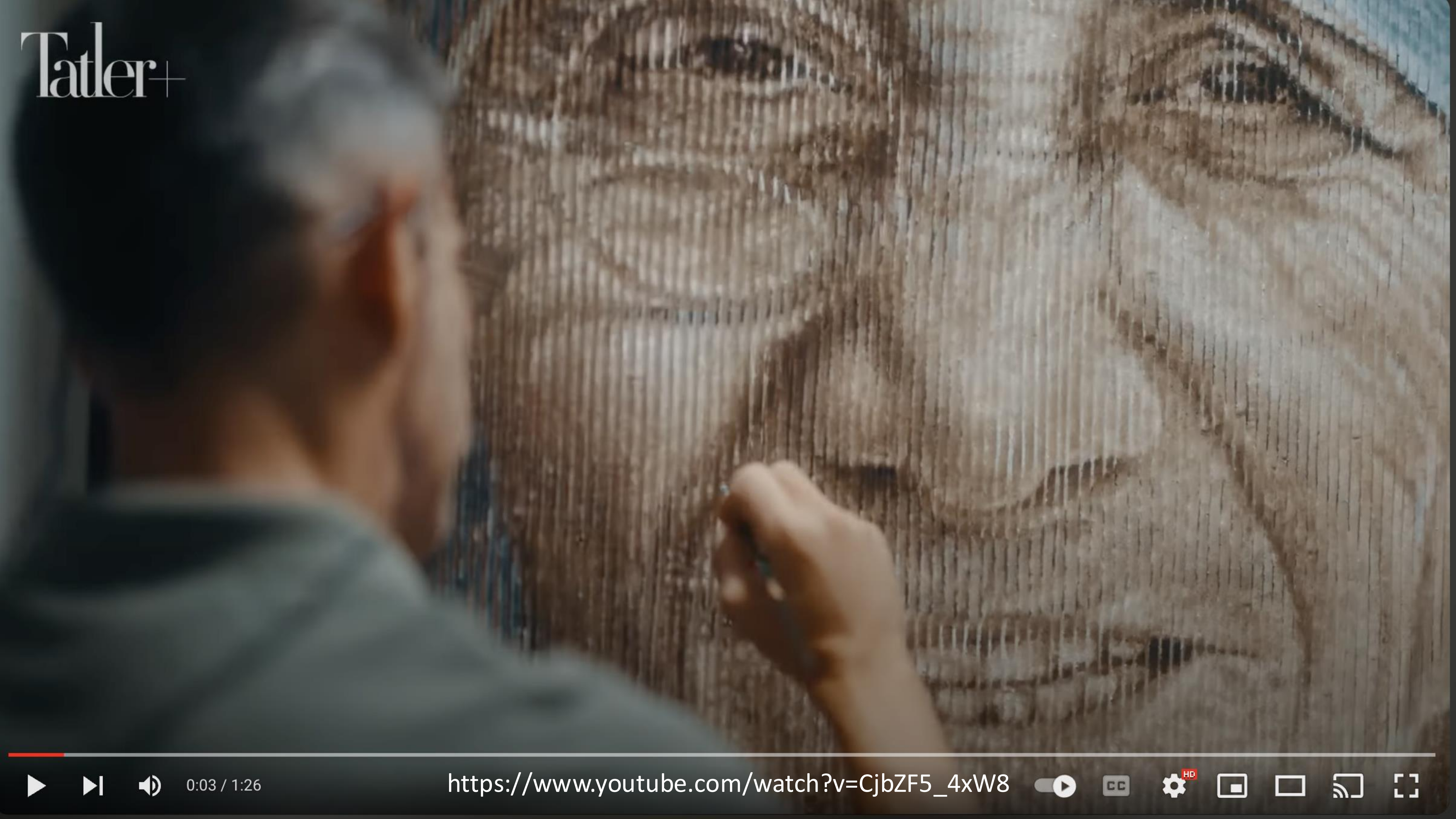
$$-5 = -(10^1 - 5) = -(10\text{'s comp. } 5)$$

10^0	–(10's comp.)	10^0
0		0
1		–9
2		–8
3		–7
4		–6
5	→	–5
6		–4
7		–3
8		–2
9		–1

10^0	$-$(10's comp.)	10^0
0		0
1		-9
2		-8
3		-7
4		-6
5	←	-5
6		-4
7		-3
8		-2
9		-1

10^0		10^0
0	Eqaul Share Base-10 Signed 10’s comp.	0
1		–9
2		–8
3		–7
4		–6
5		–5
6		–4
7		–3
8		–2
9		–1

10^0		10^0
0	Base-10 Signed 10's comp.	see 0 interpret 0
1		see 1 interpret 1
2		see 2 interpret 2
3		see 3 interpret 3
4		see 4 interpret 4
$5 \rightarrow -5$		see 5 interpret –5
$6 \rightarrow -4$		see 6 interpret –4
$7 \rightarrow -3$		see 7 interpret –3
$8 \rightarrow -2$		see 8 interpret –2
$9 \rightarrow -1$		see 9 interpret –1



https://www.youtube.com/watch?v=CjbZF5_4xW8

SIGNED 10’s COMP. Base-10

Signed Radix Complement Base-10						
10^{n-1}	10^{n-2}	10^{n-3}	…	10^2	10^1	10^0

$0 <=$ Positive Numbers $<= (10^n - 1) \div 2$

Nothing to do!

Surface == Meaning

Signed Radix Complement Base-10						
10^{n-1}	10^{n-2}	10^{n-3}	…	10^2	10^1	10^0

$(10^n - 1) \div 2 + 1 <=$	Negative Numbers	$<= (10^n - 1)$

Although we see positive numbers, we must interpret them negative!

How?

Surface != Meaning

Signed Radix Complement Base-10						
10^{n-1}	10^{n-2}	10^{n-3}	…	10^2	10^1	10^0
$(10^n - 1) \div 2 + 1 <=$		Negative Numbers		$<= (10^n - 1)$		

– (10's comp. of the number we see)

Signed Radix Complement Base-10						
10^6	10^5	10^4	10^3	10^2	10^1	10^0
5	8	0	5	0	7	4

$n = 7$

$0 <=$ Positive Numbers $<= (10^7 - 1) \div 2$

$(10^7 - 1) \div 2 + 1 <=$ Negative Numbers $<= (10^7 - 1)$

Signed Radix Complement Base-10						
10^6	10^5	10^4	10^3	10^2	10^1	10^0
5	8	0	5	0	7	4

$n = 7$

$(10^7 - 1) \div 2 = 9{,}999{,}999 \div 2 = 4{,}999{,}999$

$5{,}805{,}074 > 4{,}999{,}999$

This number must be interpreted negative!

Signed Radix Complement Base-10						
10^6	10^5	10^4	10^3	10^2	10^1	10^0
5	8	0	5	0	7	4

= – (10's comp. (5,805,074))

Signed Radix Complement Base-10						
10^6	10^5	10^4	10^3	10^2	10^1	10^0
5	8	0	5	0	7	4

= – (10's comp. (5,805,074))

= – (9's comp. (5,805,074) + 1)

Signed Radix Complement Base-10						
10^6	10^5	10^4	10^3	10^2	10^1	10^0
5	8	0	5	0	7	4

= – (10's comp. (5,805,074))

= – (9's comp. (5,805,074) + 1)

= – (9,999,999 – 5,805,074 + 1)

Signed Radix Complement Base-10						
10^6	10^5	10^4	10^3	10^2	10^1	10^0
5	8	0	5	0	7	4

= – (10's comp. (5,805,074))

= – (9's comp. (5,805,074) + 1)

= – (9,999,999 – 5,805,074 + 1)

= – (4,194,925 + 1)

Signed Radix Complement Base-10						
10^6	10^5	10^4	10^3	10^2	10^1	10^0
5	8	0	5	0	7	4

= – (10's comp. (5,805,074))

= – (9's comp. (5,805,074) + 1)

= – (9,999,999 – 5,805,074 + 1)

= – (4,194,925 + 1)

= – (4,194,926)

Signed Radix Complement Base-10						
10^6	10^5	10^4	10^3	10^2	10^1	10^0
X	X	X	X	X	X	X

– 3,450,256

Signed Radix Complement Base-10						
10^6	10^5	10^4	10^3	10^2	10^1	10^0
X	X	X	X	X	X	X

$-\ 3,450,256$

We must find its positive complement to represent it!

How?

Signed Radix Complement Base-10						
10^6	10^5	10^4	10^3	10^2	10^1	10^0

$$- 3{,}450{,}256$$
$$= 10\text{'s comp } (3{,}450{,}256)$$

Signed Radix Complement Base-10						
10^6	10^5	10^4	10^3	10^2	10^1	10^0

$-$ 3,450,256
= 10's comp (3,450,256)
= 9's comp (3,450,256) + 1

Signed Radix Complement Base-10						
10^6	10^5	10^4	10^3	10^2	10^1	10^0

$$-\ 3{,}450{,}256$$
$$=\ 10\text{'s comp}\ (3{,}450{,}256)$$
$$=\ 9\text{'s comp}\ (3{,}450{,}256)\ +\ 1$$
$$=\ 9{,}999{,}999 - 3{,}450{,}256\ +\ 1$$

Signed Radix Complement Base-10						
10^6	10^5	10^4	10^3	10^2	10^1	10^0

$$
\begin{aligned}
&- 3{,}450{,}256 \\
&= 10\text{'s comp } (3{,}450{,}256) \\
&= 9\text{'s comp } (3{,}450{,}256) + 1 \\
&= 9{,}999{,}999 - 3{,}450{,}256 + 1 \\
&= 6{,}549{,}743 + 1
\end{aligned}
$$

Signed Radix Complement Base-10						
10^6	10^5	10^4	10^3	10^2	10^1	10^0
6	5	4	9	7	4	4

– 3,450,256

= 10's comp (3,450,256)

= 9's comp (3,450,256) + 1

= 9,999,999 – 3,450,256 + 1

= 6,549,743 + 1

= 6,549,744

SIGNED 16's COMP. Base-16

<table>
<tr><td rowspan="2">Base-16</td><td>1</td><td>B</td><td>F</td><td>2</td><td>B</td><td>4</td></tr>
<tr><td colspan="6">Is this a negative number or positive?</td></tr>
</table>

Base-16	1	B	F	2	B	4
	How many positions? n = 6					

Base-16	1	B	F	2	B	4
	What is the maximum positive number? $(16^6 - 1) \div 2 = (8{,}388{,}607)_{10} = (7FF{,}FFF)_{16}$					

Base-16	1	B	F	2	B	4
	This number is less than $(7FF{,}FFF)_{16}$ $(+1BF{,}2B4)_{16}$					

Base-16	E	B	F	2	B	4
	Is this a negative number or positive?					

<table>
<tr><td rowspan="2">Base-16</td><td>E</td><td>B</td><td>F</td><td>2</td><td>B</td><td>4</td></tr>
<tr><td colspan="6">How many positions?
n = 6</td></tr>
</table>

Base-16	E	B	F	2	B	4
	What is the maximum positive number? $(16^6-1)\div 2 = (8,388,608)_{10} = (7FF,FFF)_{16}$					

Base-16	E	B	F	2	B	4
	This number is greater than $(7FF{,}FFF)_{16}$ The number is negative, but what is the number?					

<table>
<tr><td rowspan="2">Base-16</td><td>E</td><td>B</td><td>F</td><td>2</td><td>B</td><td>4</td></tr>
<tr><td colspan="6">= – (16's comp. (EBF,2B4))</td></tr>
</table>

<table>
<tr><td rowspan="2">Base-16</td><td>E</td><td>B</td><td>F</td><td>2</td><td>B</td><td>4</td></tr>
<tr><td colspan="6">= – (16’s comp. (EBF,2B4))
= – (15’s comp. (EBF,2B4) + 1)</td></tr>
</table>

<table>
<tr><td rowspan="2">Base-16</td><td>E</td><td>B</td><td>F</td><td>2</td><td>B</td><td>4</td></tr>
<tr><td colspan="6">= – (16’s comp. (EBF,2B4))
= – (15’s comp. (EBF,2B4) + 1)
= – (FFF,FFF – EBF2B4 + 1)</td></tr>
</table>

<table>
<tr><td rowspan="2">Base-16</td><td>E</td><td>B</td><td>F</td><td>2</td><td>B</td><td>4</td></tr>
<tr><td colspan="6">= − (16’s comp. (EBF,2B4))
= − (15’s comp. (EBF,2B4) + 1)
= − (FFF,FFF − EBF2B4 + 1)
= − (140,D4B + 1)</td></tr>
</table>

<table>
<tr><td rowspan="2">Base-16</td><td>E</td><td>B</td><td>F</td><td>2</td><td>B</td><td>4</td></tr>
<tr><td colspan="6">= – (16’s comp. (EBF,2B4))
= – (15’s comp. (EBF,2B4) + 1)
= – (FFF,FFF – EBF2B4 + 1)
= – (140,D4B + 1)
= – (140,D4C)</td></tr>
</table>

Base-16	X	X	X	X	X	X
	– (140,D4C)					

<table>
<tr><td rowspan="2">Base-16</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr>
<tr><td colspan="6">– 140,D4C
= 16’s comp(140,D4C)
= 15’s comp(140,D4C) + 1
= FFF,FFF – 140,D4C + 1
= EBF,2B3 + 1
= EBF,2B4</td></tr>
</table>

<table>
<tr><td rowspan="2">Base-16</td><td>E</td><td>B</td><td>F</td><td>2</td><td>B</td><td>4</td></tr>
<tr><td colspan="6">– 140,D4C
= 16’s comp(140,D4C)
= 15’s comp(140,D4C) + 1
= FFF,FFF – 140,D4C + 1
= EBF,2B3 + 1
= EBF,2B4</td></tr>
</table>

SIGNED r's COMP. Base-r

Base-r in Radix Complement

r^{n-1}	r^{n-2}	r^{n-3}	…	r^2	r^1	r^0

$0 <=$ Positive Numbers $<= (r^n - 1) \div 2$

Nothing to do!

Base-2: 0,111,…,111
Base-4: 1,333,…,333
Base-8: 3,777,…,777
Base-10: 4,999,…,999
Base-16: 7,FFF,…,FFF

Base-r in Radix Complement

r^{n-1}	r^{n-2}	r^{n-3}	…	r^2	r^1	r^0

$(r^n - 1) \div 2 + 1 <=$ Negative Numbers $<= (r^n - 1)$

Base-2: 1,000,…,000
Base-4: 2,000,…,000
Base-8: 4,000,…,000
Base-10: 5,000,…,000
Base-16: 8,000,…,000

We see positive number, but we interpret negative!

$= -$ (r's comp. (#)) $= -$ ((r–1)'s comp. (#) + 1)

Base-r in Radix Complement

r^{n-1}	r^{n-2}	r^{n-3}	…	r^2	r^1	r^0

$(r^n - 1) \div 2 + 1 <=$ Negative Numbers $<= (r^n - 1)$

Base-2: 1,000,…,000
Base-4: 2,000,…,000
Base-8: 4,000,…,000
Base-10: 5,000,…,000
Base-16: 8,000,…,000

The r's comp of these borderline numbers are themselves!
But in the second have → Negative numbers

We see positive number, but we interpret negative!

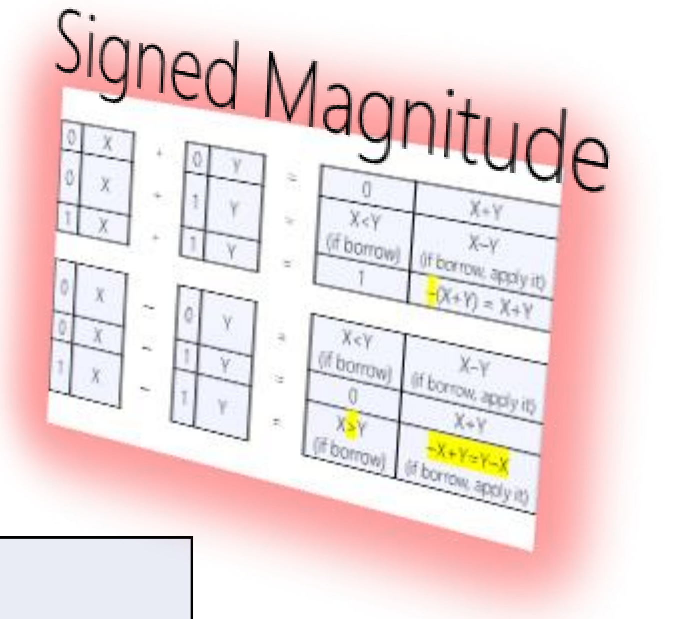
$= -$ (r's comp. (#)) $= -$ ((r–1)'s comp. (#) + 1)

Practice Base-2,3,4
1, 2, 3, 4 Positions

SIGNED 2'S COMPLEMENT BASE-2

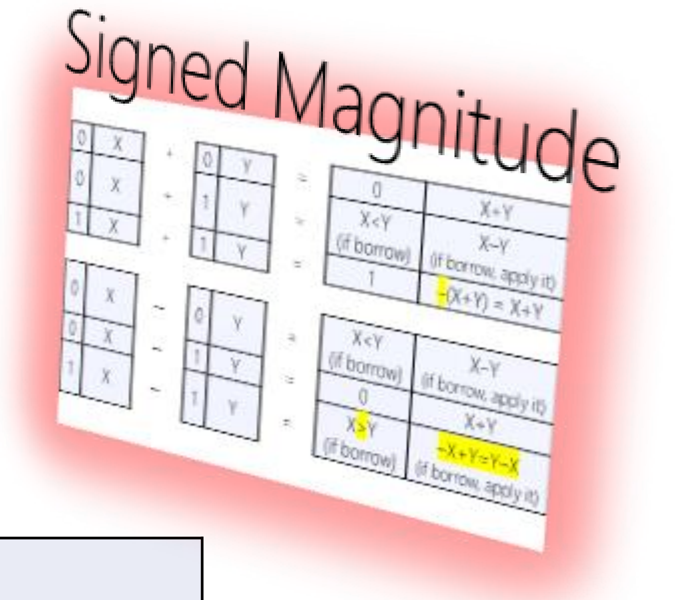
6. Show that in 2's complement binary system, the highest significant position acts like a sign (not the same) as in signed-magnitude binary system. Is this true for any radix-r number system? Justify your answer.

SIGNED r'S COMPLEMENT ARITHMETIC



X	+	Y	=	X+Y
X	–	Y	=	X+(r's comp Y)

Last Carry → Ignore



$$X + Y = X+Y$$

$$X - Y = X+(r\text{'s comp } Y)$$

One adder for both addition and subtraction!

SIGNED r'S COMPLEMENT ADDITION

+ Base-2		1	0	1	0	1	1
		0	0	1	1	1	0

+ Base-2		1	0	1	0	1	1
		$> (2^6-1)\div 2 = (011,111)_2$ $= -(2\text{'s comp}(\#))$ $= -(010101)_2$ $= -(21)_{10}$					

+ Base-2							
		0	0	1	1	1	0
		$< (2^6-1) \div 2 = (011{,}111)_2$ $= +(001110)_2$					

			1	1	1		
+ Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$+(14)_{10}$	0	0	1	1	1	0
		1	1	1	0	0	1

			1	1	1		
+ Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$+(14)_{10}$	0	0	1	1	1	0
		1	1	1	0	0	1

$> (2^6-1)\div 2 = (011{,}111)_{10}$

$= -(2\text{'s comp}(\#))$

$= -(000111)_2$

$= -(7)_{10}$

SIGNED r'S COMPLEMENT SUBTRACTION

− Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$+(8)_{10}$	0	0	1	0	0	0

–	$-(21)_{10}$	1	0	1	0	1	1
Base-2	$+(8)_{10}$	0	0	1	0	0	0
		We must convert the subtraction to addition. Eq., addition with $-(8)_{10}$!					

− Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$+(8)_{10}$	0	0	1	0	0	0
		= − (2's comp. (001000)) = − (1's comp(001000) + 1) = − (NOT(001000) + 1) = − (110111 + 1) = − (111000)					

+ Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$-(8)_{10}$	1	1	1	0	0	0

Last Carry → Ignore

	1	1	1				
+	$-(21)_{10}$	1	0	1	0	1	1
Base-2	$-(8)_{10}$	1	1	1	0	0	0
		1	0	0	0	1	1

		1	1				
+	$-(21)_{10}$	1	0	1	0	1	1
Base-2	$-(8)_{10}$	1	1	1	0	0	0
		1	0	0	0	1	1
		Done! This is the Result.					

		1	1				
+	$-(21)_{10}$	1	0	1	0	1	1
Base-2	$-(8)_{10}$	1	1	1	0	0	0
		1	0	0	0	1	1

$> 2^6 - 1 \div 2 = (011{,}111)_{10}$

The number is negative:

$= -$ (2's comp. (100011))

$= -(011101)$

		1	1				
+ Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$-(8)_{10}$	1	1	1	0	0	0
	$-(29)_{10}$	1	0	0	0	1	1

SIGNED r'S COMPLEMENT OVERFLOW

$$\boxed{X} + \boxed{Y} = \boxed{X+Y}$$

The + result of two negative numbers → positive

The + result of two positive numbers → negative

$\boxed{X} - \boxed{Y} = \boxed{X + (r\text{'s comp } Y)}$

The + result of two negative numbers → positive

The + result of two positive numbers → negative

Originally Add   After Comp. for Subtraction

X + Y

The + result of two negative numbers → positive

The + result of two positive numbers → negative

Always Check!

OVERFLOW

Example

−	$-(21)_{10}$	1	0	1	0	1	1
Base-2	$+(14)_{10}$	0	0	1	1	1	0

–	$-(21)_{10}$	1	0	1	0	1	1
Base-2	$+(14)_{10}$	0	0	1	1	1	0
		We must convert the subtraction to addition. Eq., addition with $-(14)_{10}$!					

−	$-(21)_{10}$	1	0	1	0	1	1
Base-2	$+(14)_{10}$	0	0	1	1	1	0
		= (2's comp. (001110))					
		= (1's comp(001110) + 1)					
		= (NOT(001110) + 1)					
		= (110001 + 1)					
		= (110010)					

+ Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$-(14)_{10}$	1	1	0	0	1	0

Two Negative Numbers …

Last Carry → Ignore

	1				1		
+	$-(21)_{10}$	1	0	1	0	1	1
Base-2	$-(14)_{10}$	1	1	0	0	1	0
		0	1	1	1	0	1

					1		
+ Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$-(14)_{10}$	1	1	0	0	1	0
		0	1	1	1	0	1
		$< 2^6-1 \div 2 = (011,111)_{10}$ $= (011101)_2$ Positive Number					

					1		
+ Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$-(14)_{10}$	1	1	0	0	1	0
		Overflow! Don’t rely on the result!					

Sum of two negative numbers become positive!

UNDERFLOW

$(00)_2 - (10)_2 = ?$

$(01)_2 - (10)_2 = ?$